Physical Chemistry CUME, October 30, 2020

Literature

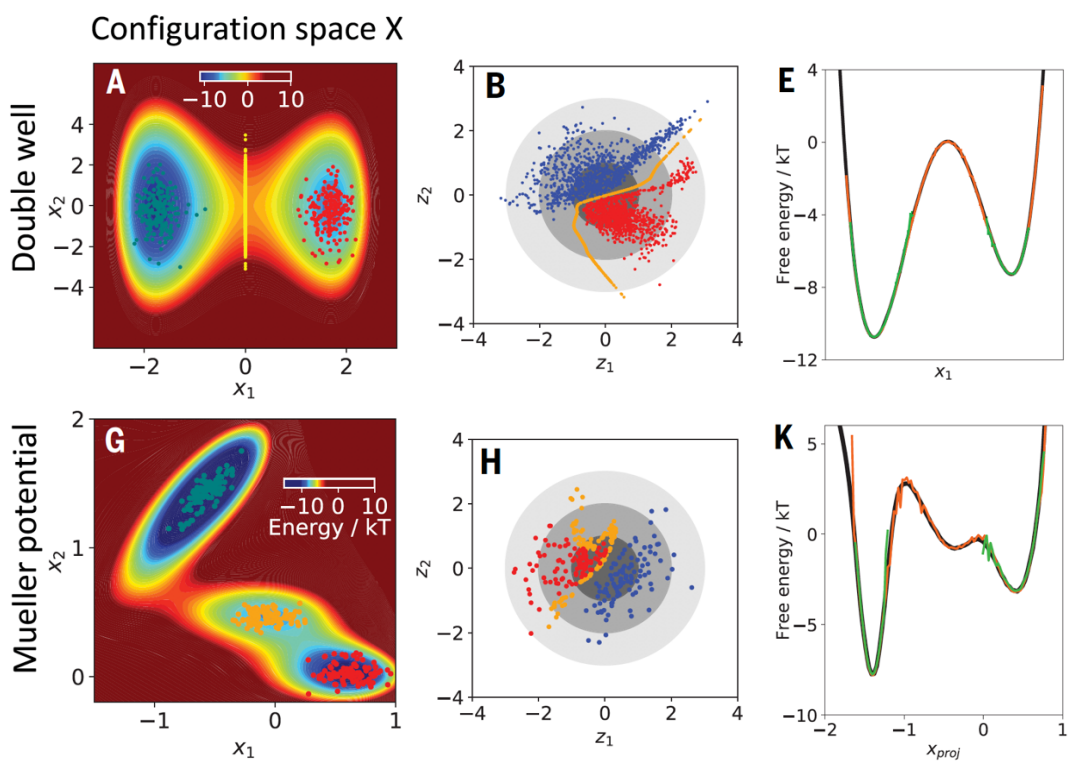
The following paper was previously announced:

Frank Noe, Simon Olsson, Jonas Kohler, and Hao Wu, "Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning", Science 365, eaaw1147 (2019)

1) The paper states that except for simple model systems, we presently have no approach to directly draw one-shot (i.e., statistically independent) samples, $\boldsymbol{x}$, from Boltzmann distributions to compute thermodynamic quantities such as free-energy differences. Therefore, one currently must rely on trajectory methods such as molecular dynamics simulations that make tiny changes to $\boldsymbol{x}$ in each step. The proposed Boltzmann generators, however, can be trained to provide unbiased, one-shot samples from a Boltzmann distribution.

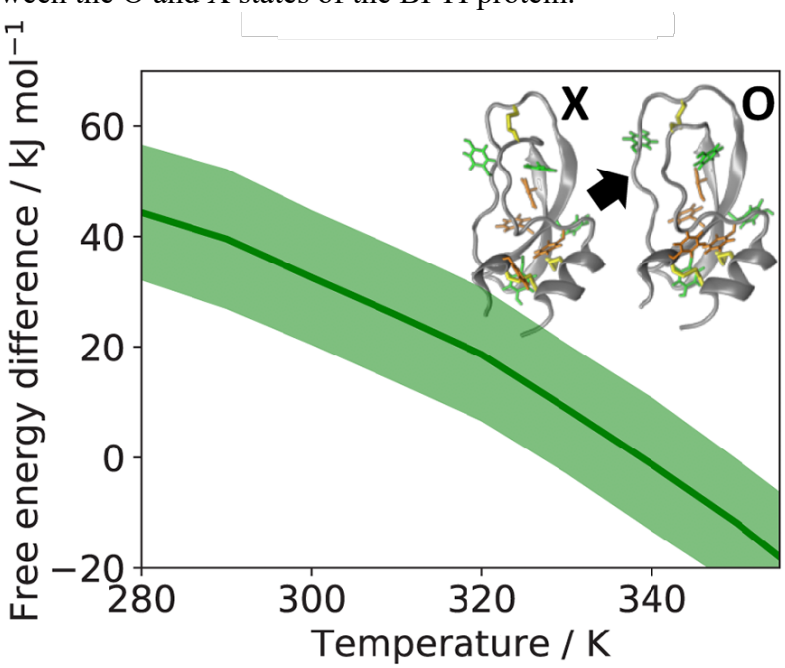
a. Explain why trajectory methods such as molecular dynamics are not efficient for estimating free energy differences for complex systems such as proteins. [20]
b. Briefly describe how estimating the free energy difference between the folded and unfolded states of a protein is different methodologically using molecular dynamics vs Boltzmann generator methods. [30]

2) Consider the Figures below from the paper. Panels A and G describe the potential energy surfaces of two 2D toy models including double-well (A) and Muller (G) potentials.



a. The blue, red, and orange samples shown in each plot (in A/B and G/H) represent different states in each potential surface. What is the key difference between the orange state in the double-well and Muller potentials? [10]
b. What do the panels B and H represent? [20]

3) Consider the Figure below from the paper showing the temperature-dependent free energy difference between the O and X states of the BPTI protein.



a. Which state has a lower average potential energy? Provide an argument based on the free energy plot above. [10]
b. Which state has a lower entropy? Provide an argument based on the free energy plot above. [10]